INFORMATION SECURITY

NAME: RUBBIYAH ABID

ROLLNO: BITF24M0002

Project Proposal: OTP/2FA Attack and Defense Simulation

Introduction

In today' s digital world, securing online accounts has become more critical than ever. **One-Time Passwords (OTP)** and **Two-Factor Authentication (2FA)** are widely used to strengthen account security by adding an extra verification step beyond the password. Despite their importance, these systems are still vulnerable to attacks such as **brute-force OTP guessing, phishing attacks, and session hijacking**.

Understanding these vulnerabilities and the corresponding defence mechanisms is crucial for cybersecurity professionals. This project provides a **controlled, hands-on simulation** of attacks against OTP/2FA systems and demonstrates how **security features like OTP expiration, rate-limiting, and account lockout** can protect user accounts. The simulation helps learners and professionals visualize the interaction between attacks and defences in real-time, improving their comprehension of authentication security.

1. Problem Statement / Project Idea

Despite the widespread adoption of 2FA, attackers continue to exploit weaknesses in OTP systems to compromise accounts. Brute-force attempts, phishing schemes, and stolen session tokens are common attack vectors that can bypass weakly implemented authentication systems.

The objective of this project is to **simulate attacks on an OTP/2FA system** safely and to implement **defence mechanisms** that prevent unauthorized access. Users will gain practical knowledge of how OTP/2FA systems work, the threats they face, and the importance of security policies like OTP expiration and account lockout.

2. Scope

The project will cover:

- **Simulating OTP/2FA attacks** such as brute-force attempts and phishing simulations.
- **Implementing security measures** including:
  - OTP expiration timers
  - Attempt rate-limiting

  - Account lockout after repeated failed attempts

- **Logging and reporting attack attempts** to analyze vulnerabilities and defense efficiency.
- **Providing a safe, educational environment** for learning about authentication security.

Limitations / Boundaries:

- The simulation focuses exclusively on OTP/2FA systems and does not cover other authentication methods like biometrics or hardware tokens.
- It is designed for **educational purposes only**; no real user accounts or live systems will be affected.
- Full integration with external applications or network services is not included.

Conclusion

This project provides a practical, hands-on demonstration of OTP/2FA security, showing both attack methods and corresponding defenses. It enhances understanding of authentication vulnerabilities and effective mitigation strategies. By simulating these scenarios, users can learn the importance of secure authentication mechanisms and gain insight into real-world information security practices.